

CLAREMORE, OK, USA

WMO: 722091

Lat: 36.294N

Lon: 95.479W

Elev: 725

StdP: 14.31

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53940

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.9	17.8	1.3	6.0	14.4	7.0	8.1	20.5	25.1	54.1	23.4	54.3	6.5	20	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	98.9	74.6	95.0	75.1	91.1	75.1	78.7	89.8	77.7	89.1	76.6	87.8	8.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	136.3	84.6	73.5	127.8	82.7	72.9	125.6	82.2	42.9	90.1	41.5	89.5	40.3	87.3	83.1

Extreme Annual Design Conditions

Extreme Annual WS							Extreme Annual Temperature										n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years										
Min	Max				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max										
(a)	(b)	(c)			(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)										
23.4	19.9	17.9			4.0	101.0	7.3	5.1	-1.2	104.6	-5.5	107.6	-9.7	110.5	-15.0	114.2										
			DB																							
			WB		3.1	80.1	6.8	1.8	-1.7	81.3	-5.7	82.4	-9.5	83.4	-14.4	84.6										

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.4	37.4	40.5	51.1	58.9	67.4	76.3	79.9	79.0	71.9	60.1	49.4	39.4
	DBStd	17.33	10.63	11.45	10.43	8.77	7.77	4.85	5.12	5.72	6.76	9.05	9.47	9.84
	HDD50	1342	410	304	113	23	2	0	0	0	0	18	122	350
	HDD65	3764	857	687	442	218	69	1	0	0	19	205	472	794
	CDD50	4768	18	38	148	291	541	789	928	899	658	332	105	21
	CDD65	1716	0	1	12	36	143	341	463	434	227	54	5	0
	CDH74	17313	1	11	87	314	1144	3381	5214	4571	2078	473	39	0
	CDH80	7546	0	1	13	63	298	1369	2624	2248	815	112	3	0
Wind	WSAvg	7.5	7.9	8.5	9.4	9.5	7.8	7.2	5.7	6.2	6.1	7.0	8.0	7.5
Precipitation	PrecAvg	44.6	2.2	2.1	3.4	4.6	5.3	5.2	3.7	3.3	4.8	3.6	3.7	2.5
	PrecMax	54.5	4.5	5.2	8.4	9.1	9.3	10.7	8.6	8.9	11.3	8.1	8.5	5.8
	PrecMin	30.7	0.1	0.2	0.8	1.9	2.3	1.6	1.1	0.1	1.1	1.0	0.2	0.1
	PrecStd	6.2	1.2	1.5	2.2	2.2	2.2	2.5	2.2	2.4	2.6	2.0	2.4	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.4	74.8	81.5	88.0	89.6	97.0	102.4	106.0	95.4	87.9	79.2	70.5
		MCWB	56.2	60.0	64.0	67.7	72.0	75.8	74.0	74.9	73.6	71.8	62.9	61.8
	2%	DB	64.3	69.9	75.5	81.1	86.2	91.4	99.1	99.0	90.8	82.4	72.8	63.7
		MCWB	54.0	57.4	61.8	65.2	71.7	75.4	74.6	74.5	74.0	67.7	60.0	54.6
	5%	DB	60.6	63.9	72.1	77.4	82.5	90.2	95.3	94.9	88.4	79.4	69.6	59.3
		MCWB	50.7	54.4	59.1	62.8	70.5	75.3	74.7	75.3	73.0	65.1	59.4	51.5
	10%	DB	54.5	59.4	68.0	73.2	81.0	88.0	91.2	90.7	84.5	75.1	64.1	54.8
		MCWB	46.3	50.7	56.9	61.7	69.7	74.7	75.8	74.7	71.5	64.4	55.6	48.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.2	63.4	66.7	70.4	76.0	79.4	81.6	79.9	77.3	73.7	66.9	63.1
		MCDB	65.6	70.1	78.2	81.5	84.6	91.0	90.4	91.6	88.9	84.2	74.4	67.1
	2%	WB	56.4	60.0	63.9	67.9	74.2	78.0	79.1	78.2	75.7	71.6	63.6	58.2
		MCDB	62.4	66.7	73.2	77.2	82.8	88.9	89.1	90.3	86.9	80.2	70.2	62.6
	5%	WB	51.4	56.4	61.6	66.3	72.5	76.6	77.9	77.0	74.4	69.3	60.4	53.7
		MCDB	59.0	62.9	69.4	74.3	81.1	86.8	89.1	89.1	84.9	74.3	66.4	56.4
	10%	WB	46.6	51.2	59.2	64.2	70.9	75.3	76.8	75.8	73.1	66.8	57.8	49.6
		MCDB	53.2	58.5	65.9	71.3	78.7	84.9	88.2	87.9	82.5	73.2	64.1	54.7
Mean Daily Temperature Range	5% DB	MDBR	20.6	20.7	21.4	21.8	19.7	19.3	21.0	20.5	21.6	22.7	22.1	19.4
		MCDBR	28.9	28.0	25.8	25.6	21.5	22.0	25.5	25.5	24.0	25.7	26.4	25.5
		MCWBR	18.9	17.7	14.6	14.1	10.6	8.8	7.6	7.9	9.6	12.8	16.2	16.6
	5% WB	MCDDBR	25.3	24.9	21.0	21.1	18.8	18.9	19.9	20.5	20.4	18.5	21.7	19.9
		MCWBR	18.2	17.6	13.6	13.5	10.3	8.8	8.1	8.1	9.1	11.4	15.6	16.7
Clear-Sky Solar Irradiance	taub	0.303	0.310	0.355	0.386	0.427	0.430	0.444	0.446	0.394	0.350	0.316	0.297	
	taud	2.478	2.475	2.384	2.315	2.261	2.281	2.285	2.250	2.363	2.470	2.509	2.512	
	Ebn at Noon	284	296	289	284	273	271	266	263	272	277	277	279	
	Edh at Noon	26	30	36	40	43	43	42	43	36	30	25	24	
All-Sky Solar Radiation	RadAvg	797	982	1298	1626	1806	2067	2079	1881	1574	1172	852	676	
	RadStd	76	146	75	99	167	196	128	128	125	147	101	82	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page